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Project 3

ECEN 3400

12/05/25

Project 3

Part 1 - Finite-Difference Time Domain Method for the Telegrapher's Equation:

The sample code provided solves the telegrapher's equations on a transmission line. It uses the Finite-Difference Time-Domain technique to do so. Various plots are generated and explained to observe the behavior of multiple simulation scenarios.

When the generator's impedance is set to $50\ \Omega$ the simulation is run. The plots created are as follows:

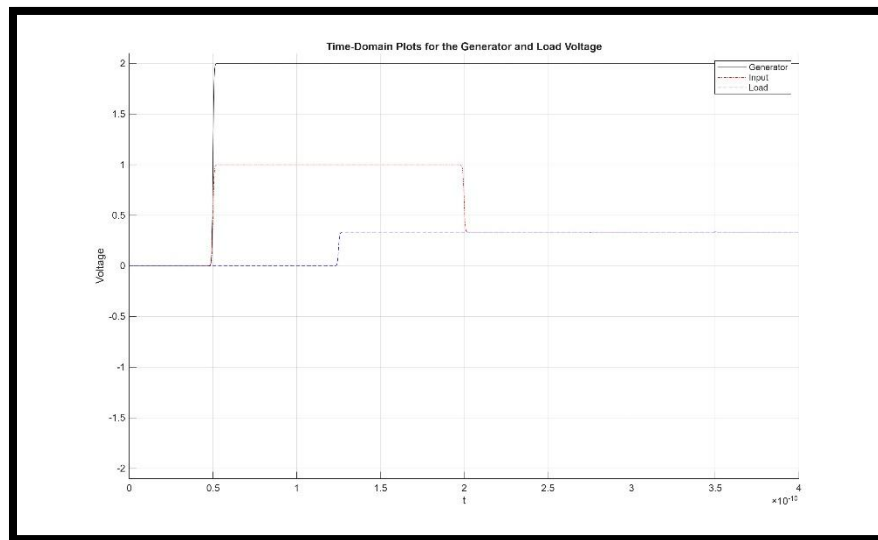


Figure 1: $50\ \Omega$ Impedance Time Domain Plot

When the generator impedance is set equal to the characteristic impedance of the transmission line, the source matches the line. In the plot, it can be seen that the initial step

launched onto the line experiences no reflection at the generator. The wave begins propagating cleanly toward the load. The source is a 2 V step feeding the $50\ \Omega$ line through a $50\ \Omega$ generator, so the initial incident voltage step has amplitude. This is clearly seen in the plot.

Now, when the wave reaches the load $10\ \Omega$ load, the impedance mismatch creates a negative reflection. This creates the step down that appears in the plot once that reflected wave returns to the line input. Since the generator is matched, that reflected wave is absorbed rather than re-reflected meaning the transient behavior will disappear.

At the end of the simulation, the transmission line achieves a flat DC state, as seen in the plot, meaning the line should behave like a wire. It looks like the input and load voltage both settle around 0.33 V.

The next simulation is run but with a $50\ \Omega$ load alongside the $50\ \Omega$ generator impedance well.

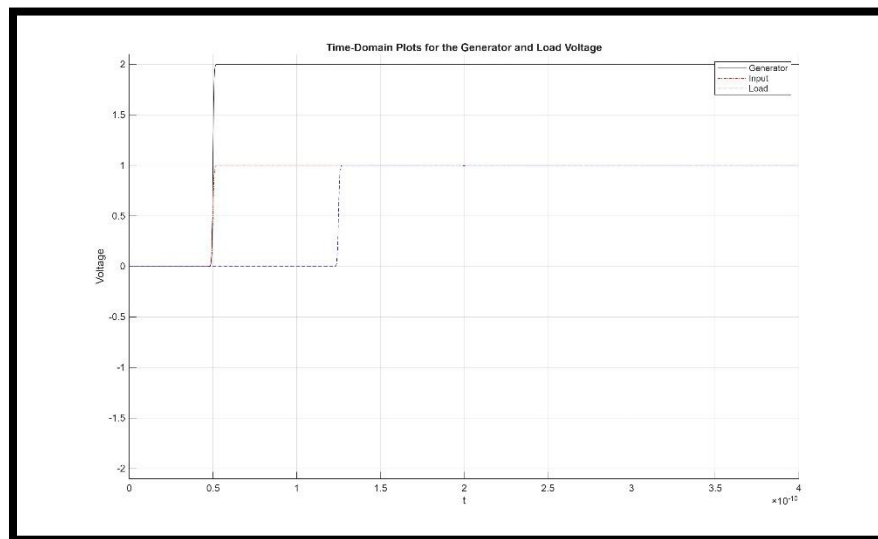


Figure 2: $50\ \Omega$ Impedance and $50\ \Omega$ Load Impedance Time Domain Plot

This time, when both the generator and load impedances are equal to the characteristic impedance of the $50\ \Omega$ line, the transmission line is fully matched. The initial 2 V step launches a 1 V wave on the line. Since the load is matched, the incident wave is completely absorbed. This is why there is a clearly visible vertical rise in the load voltage this time. The input voltage also shows no extra jump. By the end of the simulation the voltage on the line becomes spatially uniform at 1 V. This also happens to be the expected DC value of a $50\ \Omega$ - $50\ \Omega$ divider. This plot demonstrates a perfect match at both ends.

Another simulation is run next. Keeping $50\ \Omega$ for both, the generator impedance and load, a $30\ \Omega$ resistance per meter is now added too.

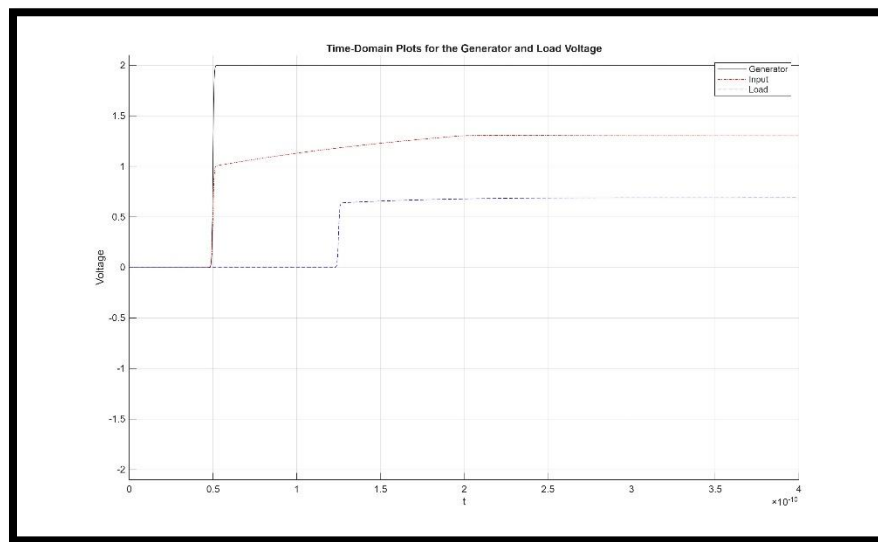


Figure 3: $50\ \Omega$ Generator and Load Impedance and $30\ \Omega/\text{m}$ R/L Time Domain Plot

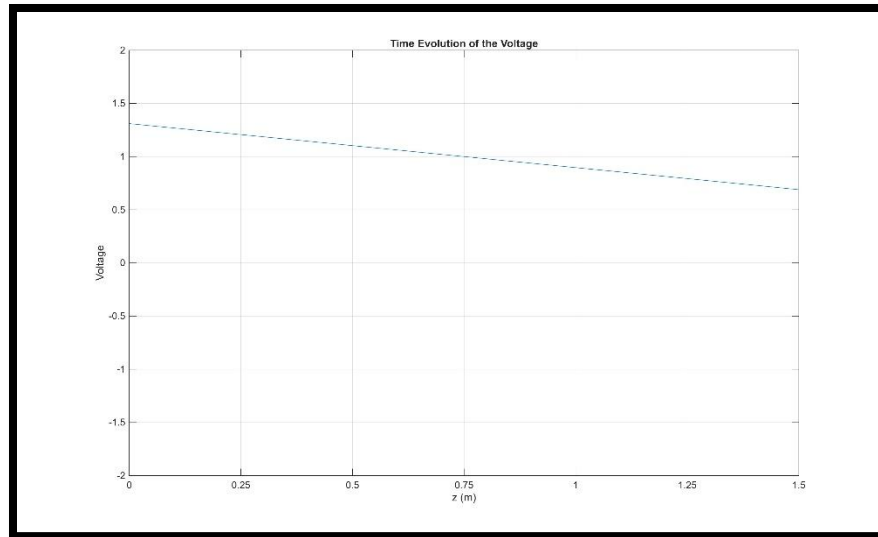


Figure 4: 50 Ω Generator and Load Impedance and 30 Ω/m R/L Final Line Potential

In the time domain plot, when the step source is turned on, there is an initial jump up to 1 V at the input. This is similar to the last simulation. This is due to the generator impedance matching the characteristic impedance. This time though, the input and load voltages do not remain flat. After the wave reaches the load, both voltages slowly increase to where they flatten out to their end values. Since the generator and load are still matched there are no reflections again. Near the end of the simulation, it reaches a steady state again.

The second plot, the final line potential, shows a steadily decreasing straight line. It starts at a higher voltage, ~ 1.4 V, and decreases to a lower voltage, ~ 0.7 V. This linear drop is what should be expected. The linear should carry a constant DC current near the end with the distributed series resistance. This is what causes this linear voltage drop. Just like the time domain plot, since the generator and load impedances are the same, there are no wave artifacts in the plot.

Adding this series resistance took the uniform behavior of the previous simulation and altered the steady-state voltage over the line. Now the load voltage has been reduced to less than 1 V with the final line potential plot illustrating the voltage drop.

The next simulation run utilizes a generator impedance of $50\ \Omega$, a load impedance of $30\ \Omega$, a resistance per meter of 0, as well as multiplying the generator by a 40 GHz cosine signal.

The plots are as follows:

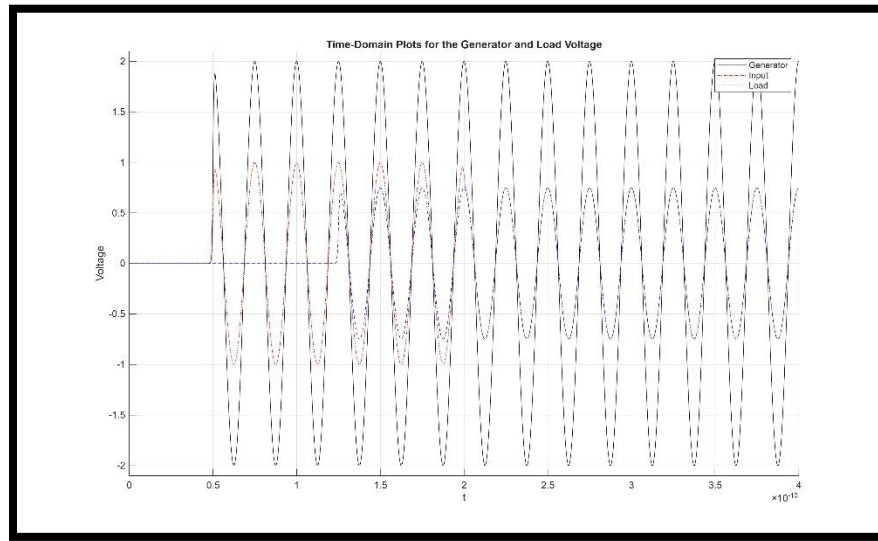


Figure 5: $50\ \Omega$ Generator, $30\ \Omega$ Load, $0\ \Omega/m$, and 40 GHz Cosine Signal Time Domain

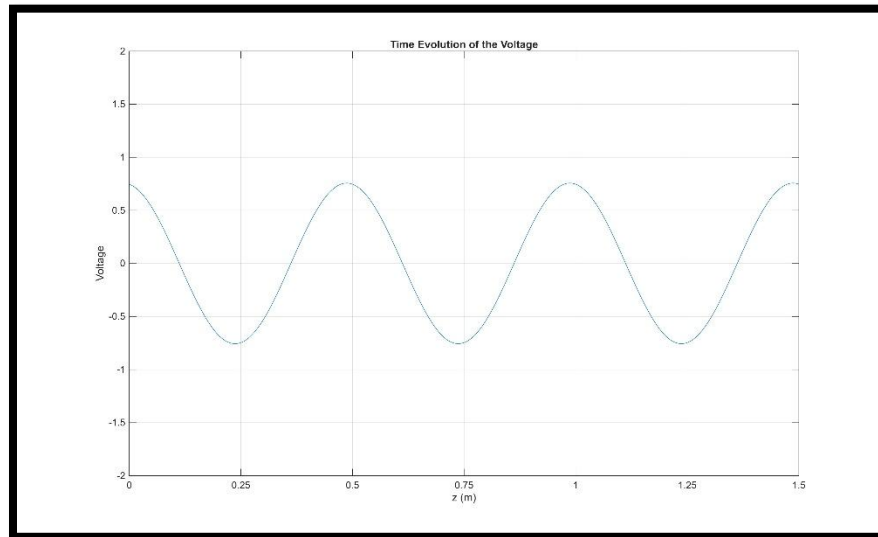


Figure 6: $50\ \Omega$ Generator, $30\ \Omega$ Load, $0\ \Omega/m$, and 40 GHz Cosine Signal Final Line

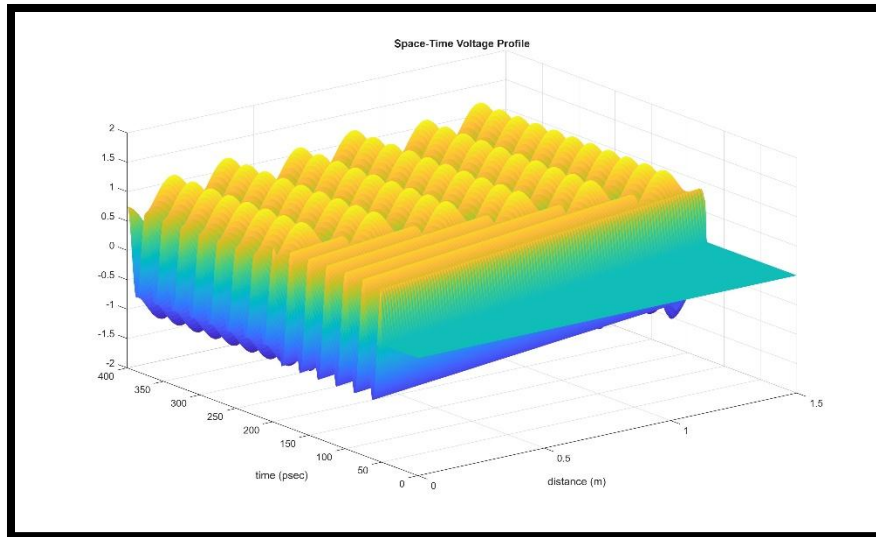


Figure 7: 50 Ω Generator, 30 Ω Load, 0 Ω/m , and 40 GHz Cosine Signal Space-Time Line

For this case, the time domain plot shows the generator voltage with a constant amplitude corresponding to the 40 GHz cosine signal applied. For the input voltage, the plot shows that a reduced amplitude compared to the generator voltage. When the wave reaches the load voltage, the amplitude is even smaller than the already decreased input voltage amplitude. It is also ever so slightly shifted too. A reflected wave returns from the load and is absorbed by the source instead of being re-reflected. The input and load voltages eventually settle into a steady sinusoidal behavior due to this effect. Besides the expected oscillations, there does not appear to be any abnormal behavior in the plot.

Looking at the space-time of the line potential, it shows the changing of sinusoidal voltage over the line. A wave travels forward from the generator to the load. When it reaches the load, a reduced amplitude wave is reflected back to the source. This causes a clear wave pattern on the line visible in both space and time.

Another experiment can be completed if the code is modified to allow for conductance per unit length. I will use G' in the Telegrapher's equation. G' is essentially the leakage between the conductors of a line. I will match the impedance of the generator and the load at $50\ \Omega$ and set the resistance per length (R') to 0. I will set my newly added term of G' to be $0.2\ \text{S/m}$. I also added a decay term in proportion to the voltage. It basically is Ampere's law with a shunt conductance. Here are the results of the simulation:

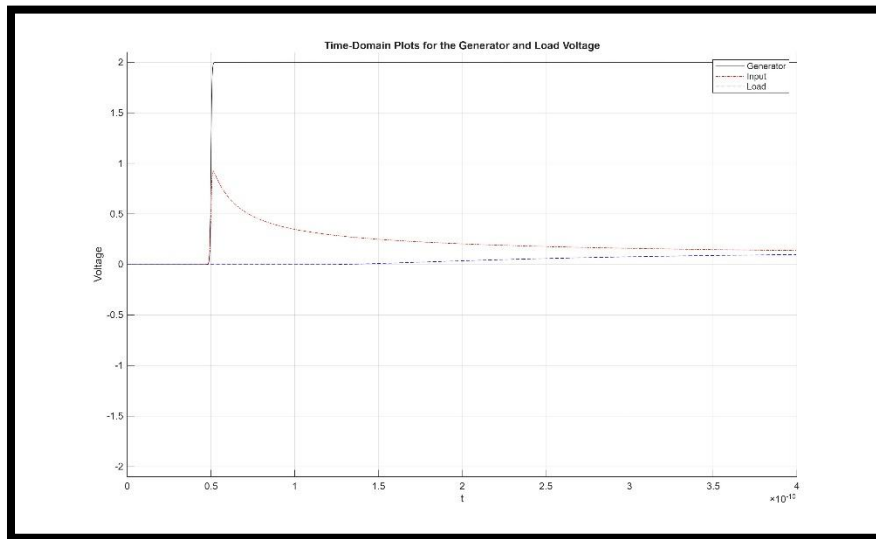


Figure 8: $50\ \Omega$ Generator, $50\ \Omega$ Load, $0\ \Omega/\text{m}$, and $0.2\ \text{S/m}$ Time Domain

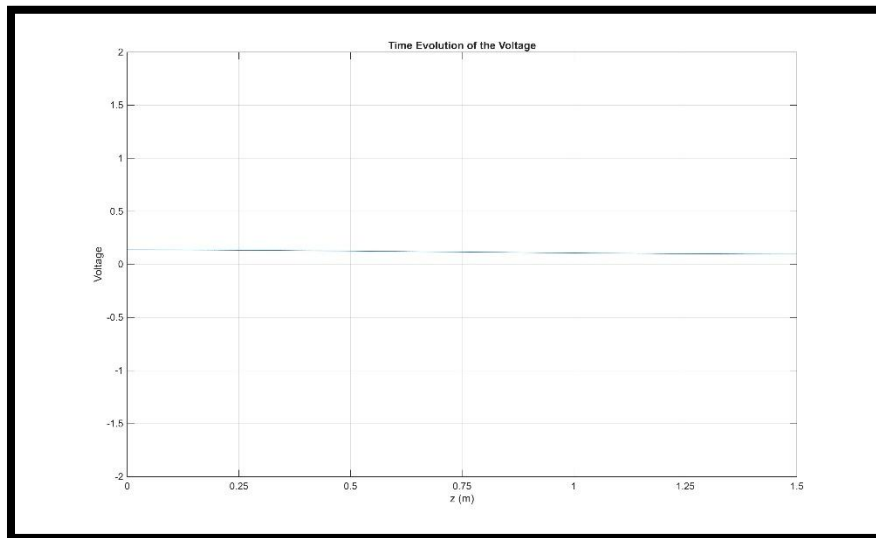


Figure 9: $50\ \Omega$ Generator, $50\ \Omega$ Load, $0\ \Omega/\text{m}$, and $0.2\ \text{S/m}$ Final Line Potential

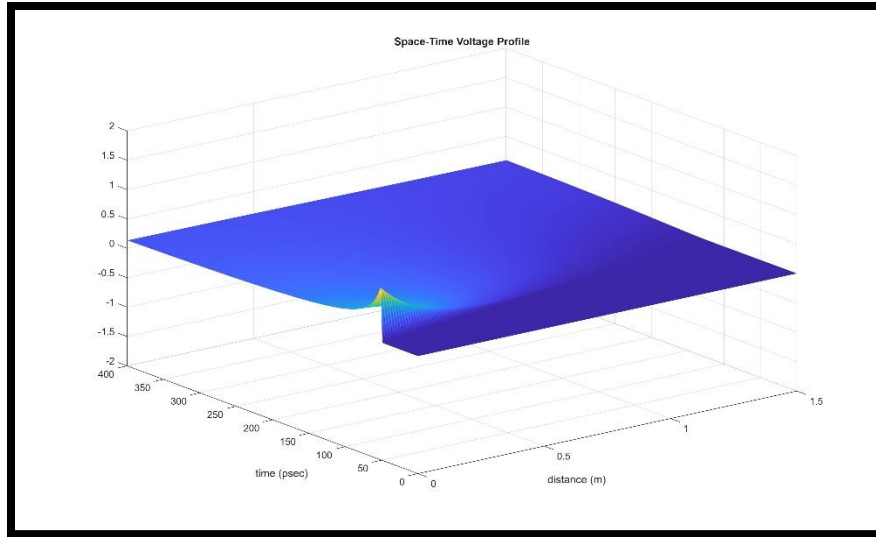


Figure 10: 50 Ω Generator, 50 Ω Load, 0 Ω/m , and 0.2 S/m Space-Time Profile

The time-domain plot reveals the effect of dielectric leakage. To start, the input voltage increases to just under 1 V. That part is expected, but it does not stay constant like observed in the second simulation (lossless). The input voltage keeps steadily decaying as time goes on. The load voltage starts at zero and barely rises. This exponential decay illustrates that shunt conductance draws current from the transmission line to the dielectric. The line never settles to a constant DC behavior. Each voltage slowly decreases as energy is constantly lost over time.

The space-time plot shows how leakage affects the wave. You can see an initial wave to start with near the generator but its amplitude keeps decreasing as the simulation keeps running. Unlike in the second simulation, where the wave keeps its shape but loses amplitude, this simulation reveals a change in shape while also losing amplitude. It eventually completely flattens out and the wave cannot be seen anywhere on the line.

The final line potential plot reveals that the voltage at the end of the simulation ($\sim t \geq 400$ ps) is nearly zero. There is a consistent, but small, decrease over the entire length. This reflects the fact that a small DC leakage current still flows due to the generator voltage, but the

overall field is essentially extinguished. This is consistent with a line whose dielectric conducts enough current to drain the wave energy before it can reach a steady DC distribution.

Part 2 - Fresnel Reflection Coefficients:

The second part of this project involves considering a plane wave illuminating an interface between two dielectrics. MATLAB code is constructed to find the Brewster angle and Critical angle for two situations. The results are as follows:

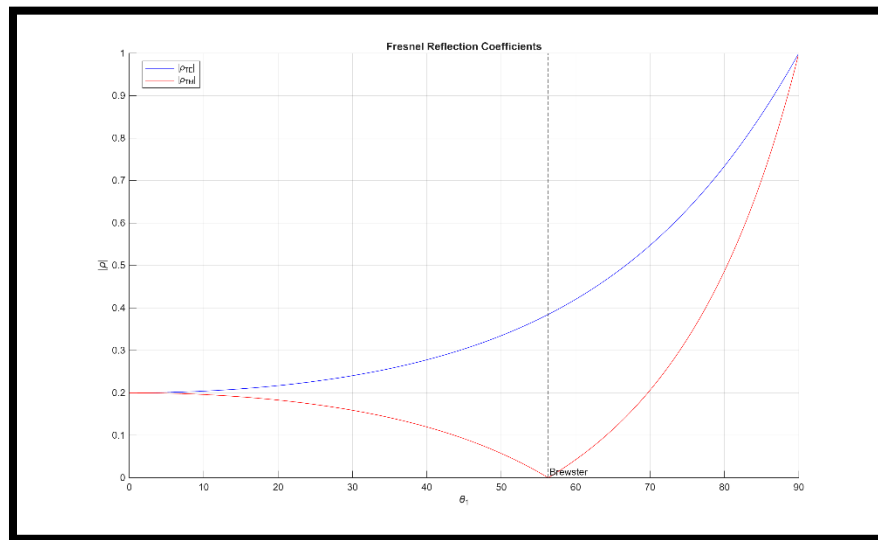


Figure 11: Magnitude of the Fresnel Reflection Coefficients for $\epsilon_{r1} = 4$ and $\epsilon_{r2} = 9$

For the first scenario with $\epsilon_{r1} = 4$ and $\epsilon_{r2} = 9$, the Brewster angle computes to $\sim 56.31^\circ$ (as calculated and printed in my MATLAB code).

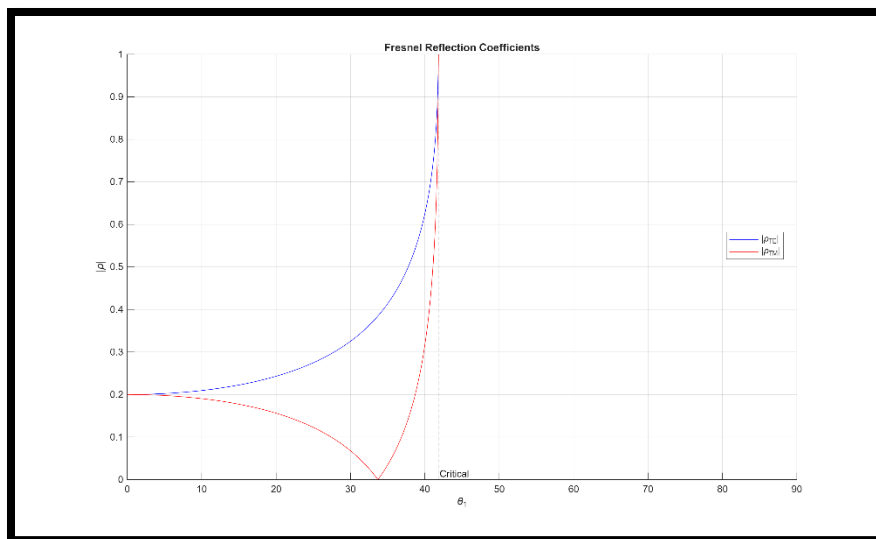


Figure 12: Magnitude of the Fresnel Reflection Coefficients for $\epsilon_{r1} = 9$ and $\epsilon_{r2} = 4$

For the second scenario with $\epsilon_{r1} = 4$ and $\epsilon_{r2} = 9$, the critical angle computes to $\sim 41.81^\circ$ (as calculated and printed in my MATLAB code). The MATLAB code can be found below.

```

%Leo Marcinkus
%ECEN 3400
%Project 3

clear;
clc;
close all

eta0 = 376.730313668; %Free space impedance in ohms
mu_r1 = 1; mu_r2 = 1; %Assuming Non-magnetic

thetal_deg = linspace(0,89.9,1000); %Angle sweep from 0 to 90 degrees
thetal = deg2rad(thetal_deg); %Convert angle sweep to radians

%% Part 2 first problem:
eps_r1 = 4; %Permittivity of region 1
eps_r2 = 9; %Permittivity of region 2

%Refractive indices
n1 = sqrt(eps_r1 * mu_r1);
n2 = sqrt(eps_r2 * mu_r2);

%Impedances
eta1 = eta0 * sqrt(mu_r1 / eps_r1);
eta2 = eta0 * sqrt(mu_r2 / eps_r2);

%Snell's law
sin_t2 = (n1/n2) .* sin(thetal); %Compute sin(theta2)
theta2 = asin(sin_t2); %Convert to actual angle

%Cosines for TE/TM
c1 = cos(thetal);
c2 = cos(theta2);

%Fresnel reflection coefficient (TE)
rho_TE = (eta2.*c1 - eta1.*c2) ./ (eta2.*c1 + eta1.*c2);

%Fresnel reflection coefficient (TM)
rho_TM = (eta2.*c2 - eta1.*c1) ./ (eta2.*c2 + eta1.*c1);

%Brewster angle formula (only applies to TM polarization)
thetaB_rad = atan(n2/n1); %Analytical Brewster angle
thetaB_deg = rad2deg(thetaB_rad);

%Numerical Brewster angle
 [~, idxB] = min(abs(rho_TM));
thetaB_num_deg = thetal_deg(idxB);

%Print the angle results
fprintf('(eps1=4, eps2=9): Brewster angle = %.2f deg\n',thetaB_deg);

```

```

%% Plotting
figure(1);
clf;
hold on
plot(theta1_deg, abs(rho_TE), 'b', 'DisplayName', '\rho_{TE}');
plot(theta1_deg, abs(rho_TM), 'r', 'DisplayName', '\rho_{TM}');

% Draw vertical line at Brewster angle
xline(thetaB_deg, 'k--', 'Brewster', 'LabelOrientation', 'horizontal', 'LabelVerticalAlignment', 'bottom', 'HandleVisibility', 'off');

grid on
xlabel('\theta_1')
ylabel('\rho')
title('Fresnel Reflection Coefficients')
legend('Location', 'best')

print(gcf, 'Part1', '-dpng', '-r300');

%% Part 2 second problem:

eps_r1 = 9; %Permittivity changes
eps_r2 = 4; %Region 2 now has the smaller permittivity

%Refractive indices
n1 = sqrt(eps_r1 * mu_r1);
n2 = sqrt(eps_r2 * mu_r2);

%Impedances
eta1 = eta0 * sqrt(mu_r1 / eps_r1);
eta2 = eta0 * sqrt(mu_r2 / eps_r2);

%Snell's Law
sin_t2 = (n1/n2) .* sin(theta1);
theta2 = asin(sin_t2);

c1 = cos(theta1);
c2 = cos(theta2);

%Fresnel reflection coefficients
rho_TE = (eta2.*c1 - eta1.*c2) ./ (eta2.*c1 + eta1.*c2);
rho_TM = (eta2.*c2 - eta1.*c1) ./ (eta2.*c2 + eta1.*c1);

%Compute critical angle for total internal reflection
thetaC_rad = asin(n2/n1);
thetaC_deg = rad2deg(thetaC_rad);

%Prints the critical angle
fprintf('(eps1=9, eps2=4): Critical angle = %.2f deg\n', thetaC_deg);

```

```

%% Plotting
figure(2);
clf;
hold on
plot(theta1_deg, abs(rho_TE), 'b', 'DisplayName', '\rho_{TE}');
plot(theta1_deg, abs(rho_TM), 'r', 'DisplayName', '\rho_{TM}');

%Vertical line for critical angle
xline(thetaC_deg, 'k--', 'Critical', 'LabelOrientation', 'horizontal',
'LabelVerticalAlignment', 'bottom', 'HandleVisibility', 'off');

grid on
xlabel('\theta_1');
ylabel('|\rho|');
title('Fresnel Reflection Coefficients');
legend('Location', 'best');
print(gcf, 'Part2', '-dpng', '-r300');

```